

Unit 6, Lesson 9: Exploring Systems of Linear Inequalities



Benchmark

MA.912.AR.9.4 Graph the solution set of a system of two-variable linear inequalities.



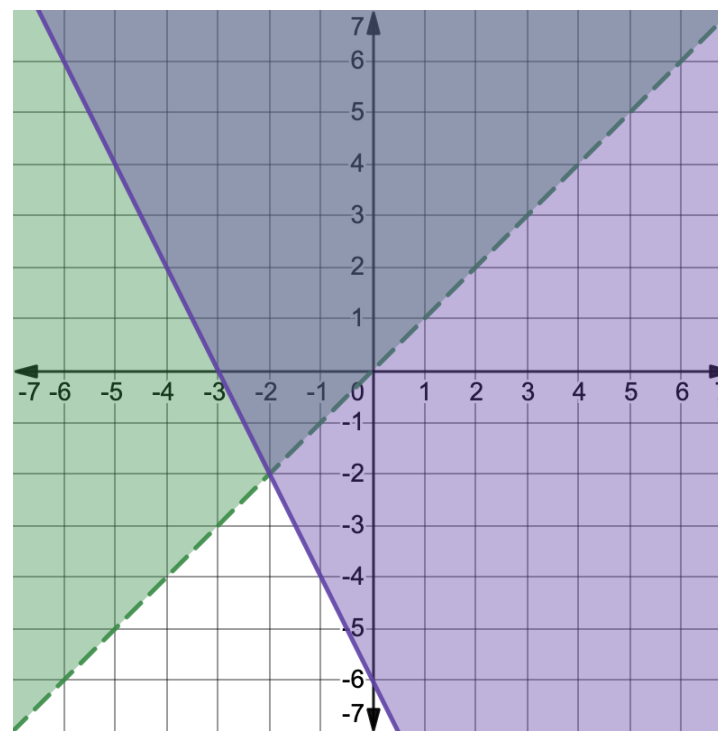
Learning Target

- I can graph the solution set of a system of inequalities.



6.9.1 Warm-Up: Notice and Wonder

1. A graph is shown on the coordinate grid.

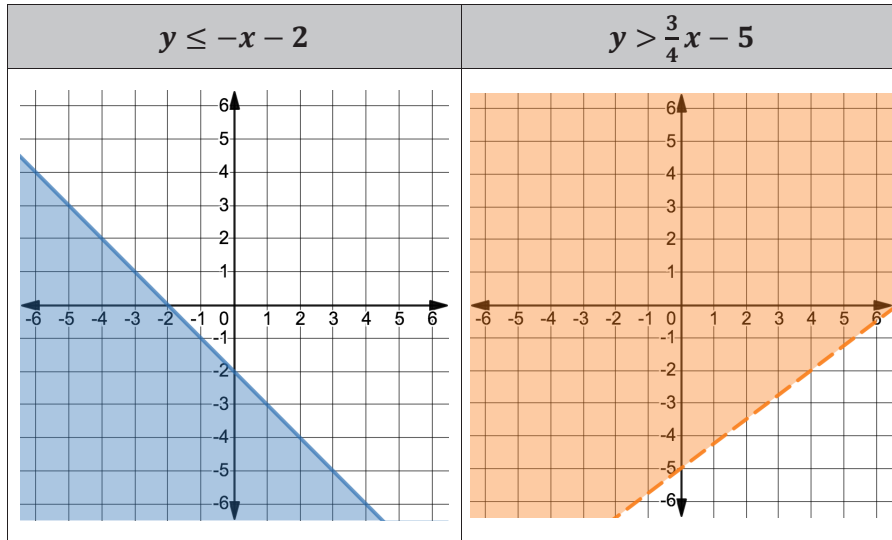


- a. What do you notice?
- b. What do you wonder?



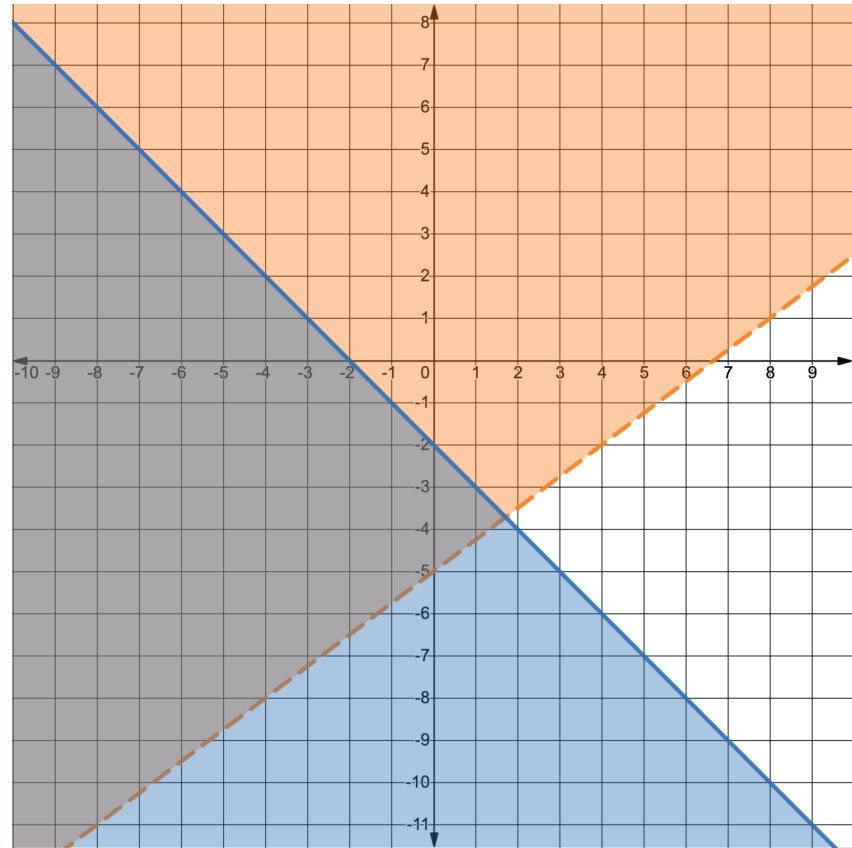
6.9.2 Exploration: Representing the Solution Set of a System of Linear Inequalities

1. The solution sets of two inequalities are graphed.



- With your partner, find a point that is in the solution set of both of the inequalities.
- Mark the point on both graphs.
- Explain how you determined a point that was in both solution sets.

2. The solution sets of both inequalities from the previous question are graphed on the coordinate grid shown.



- a. Plot each point given in the table. Then, determine whether each point is a solution to the inequalities.

Point	Solution to $y \leq -x - 2$?	Solution to $y > \frac{3}{4}x - 5$?
(4, -9)	☐ Yes ☐ No	☐ Yes ☐ No
(9, 1)	☐ Yes ☐ No	☐ Yes ☐ No
(-8, -7)	☐ Yes ☐ No	☐ Yes ☐ No
(8, 3)	☐ Yes ☐ No	☐ Yes ☐ No

- b. What do you notice about the location on the graph of the point that satisfies both inequalities?

- c. Discuss with your partner how to determine from the graph the set of points that satisfies **both** inequalities. Summarize your discussion.



6.9.3 Guided Instruction: Solution Sets for a System of Inequalities

A **system of inequalities** is a set of two or more inequalities. The solution set includes the values for the variables that make all inequalities in the system true.

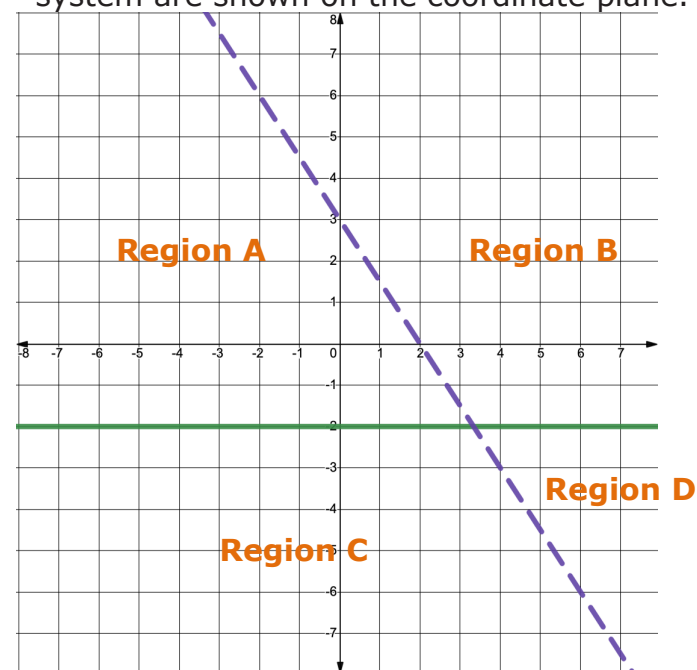
Similar to systems of linear equations, for an ordered pair to be a solution to a **system of inequalities**, the x - and y -values must satisfy both inequalities.



The solution set of a **system of inequalities** is the **intersection** of the solution sets of the individual inequalities.

1. Consider the system of linear inequalities, $\begin{cases} y \geq -2 \\ 3x + 2y > 6 \end{cases}$

- a. The boundary lines of both inequalities in the system are shown on the coordinate plane.



- b. The boundary lines separate the plane into four regions, labeled A-D. Discuss with your partner which region includes the solution set of the system of inequalities.
- c. Shade the region that includes the solution set of the system of inequalities.
- d. Select a test point in the shaded region of the solution set.
- e. Substitute the x - and y -values of the test point to show that the ordered pair satisfies both inequalities.

$y \geq -2$	$3x + 2y > 6$

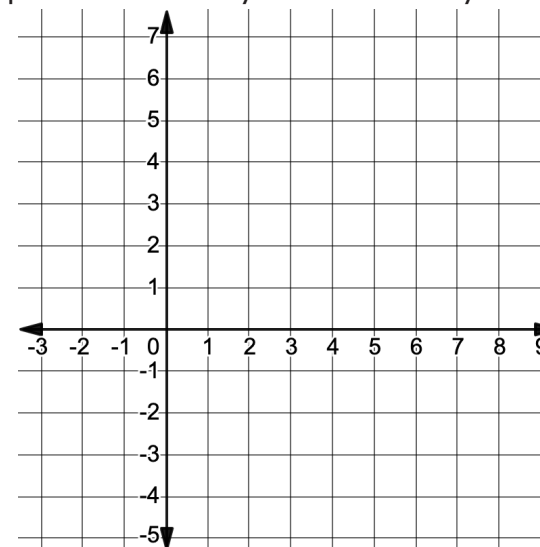
- f. Mireya says that the point $(5, -2)$ is a solution to the system of inequalities. Explain whether or not you agree.



6.9.4 Guided Practice: Graphing the Solution Set of a System of Inequalities

1. Consider the system of inequalities, $\begin{cases} y - 6 < 0.5(x - 2) \\ y + 1 \geq -2(x - 5) \end{cases}$.

- a. Graph the boundary lines of the system.



- b. Use substitution to determine whether the given ordered pairs are solutions to the system.

	$y - 6 < 0.5(x - 2)$	$y + 1 \geq -2(x - 5)$	Solution to the system?
$(8, 5)$			☐ Yes ☐ No
$(0, -3)$			☐ Yes ☐ No

- c. Complete the graph by shading the region included in the solution set of the system.





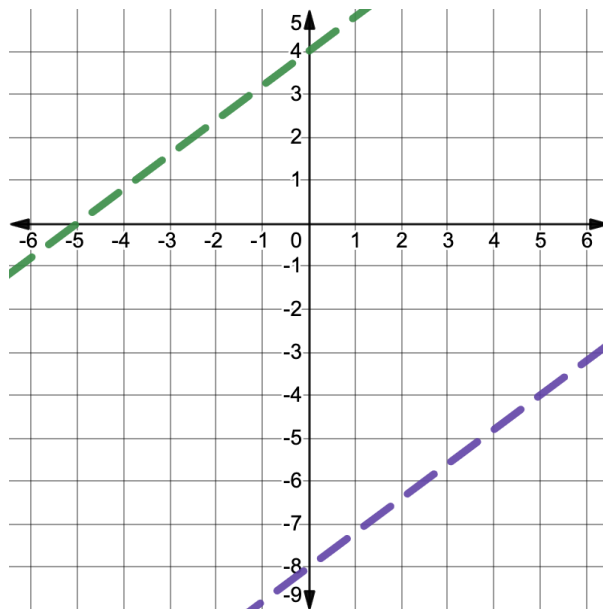
6.9.5 Collaboration: Special Systems

Work with your partner to complete the following.

1. Consider the system of inequalities.

$$\begin{cases} -4x + 5y > 20 \\ -4x + 5y < -40 \end{cases}$$

The graphs of the boundary lines of each inequality in the system are shown.



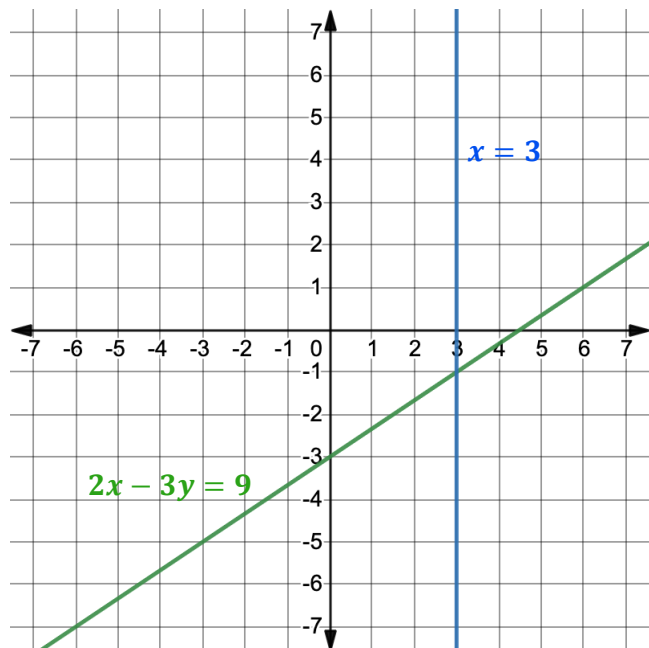
- a. What do you notice about the boundary lines in the system?

- b. Work with your partner to describe the solution set of this system of inequalities.
- c. Discuss with your partner how the solution set of the system $\begin{cases} -4x + 5y > 20 \\ -4x + 5y > -40 \end{cases}$ would change if the **green inequality symbol** were changed as shown. Summarize your discussion.



6.9.6 Your Turn!

1. The lines $x = 3$ and $2x - 3y = 9$ are graphed on the coordinate plane.



Shade the graph to represent the solution set of the system of inequalities $\begin{cases} 2x - 3y \geq 9 \\ x \geq 3 \end{cases}$.



6.9.7 Wrap-Up: Ticket Out the Door

1. Graph the solution set of the system of inequalities.

$$\begin{cases} y < -\frac{1}{4}x + 7 \\ y \geq \frac{3}{2}x - 1 \end{cases}$$

